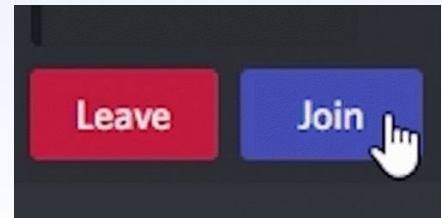




Introduction to Deep Learning: From Foundations to Frontiers.

Beyond the "Intro" –
Building Intelligence



Shreyash Tiwari

Senior Data Scientist | Educator



- Have over 7 years of experience in data science and education. Currently working with **NFL** as consultant.
- Did my Bachelor's and Master's in Economics at **IIT Kanpur**, with minors in AI/ML and Applied Maths.
- Former Data Scientist at **Amazon**, where i help build impactful ML systems and statistical models.
- Taught *Probability and Statistics* here at NST in 5th sem
- Have mentored 2000+ learners across colleges and corporates in India.

Ashwin Tewary

Data Scientist | Researcher & Educator



I specialize in **Computational Data Science**, leveraging advanced analytical techniques to drive **data-informed policy** as a **Public Sector Data Scientist** for the Australian Government.

A Dalyell Scholar from the University of Sydney, I am currently advancing my expertise in Data Science and AI at IIT Bombay.

My focus is on bridging rigorous academic foundations with practical deployment, advocating for **ethical and impactful DS/AI solutions** that **solve complex, real-world problems** and maximize public sector efficiency.

Sparsh Bansal

Data Engineer | Educator



With a industry experience at **Microchip Technology**, and education from **NIT Kurukshetra**, I work at the intersection of strong technical foundations and practical deployment.

I specialize in Applied AI and Machine Learning, focusing on turning raw data into meaningful insights and **real-world ML solutions**.

My focus is on helping students understand how AI models actually work, apply them responsibly, and **build industry-relevant projects** that solve real problems—not just run code.

Welcome Back: The Probability Connection



Probability & Statistics: Measuring Uncertainty

Recap: In Probability & Statistics, we learned to answer the question: 'Given the data, what is the likelihood?'

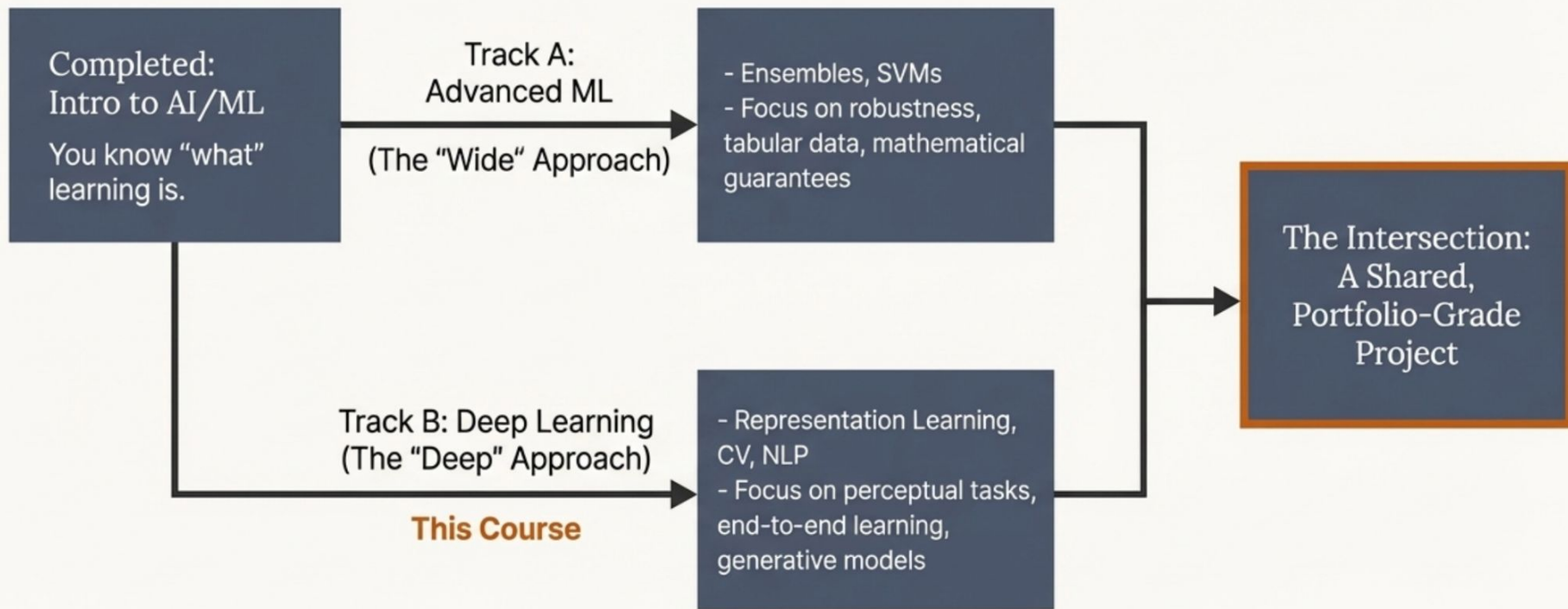


Deep Learning: Modelling Complexity

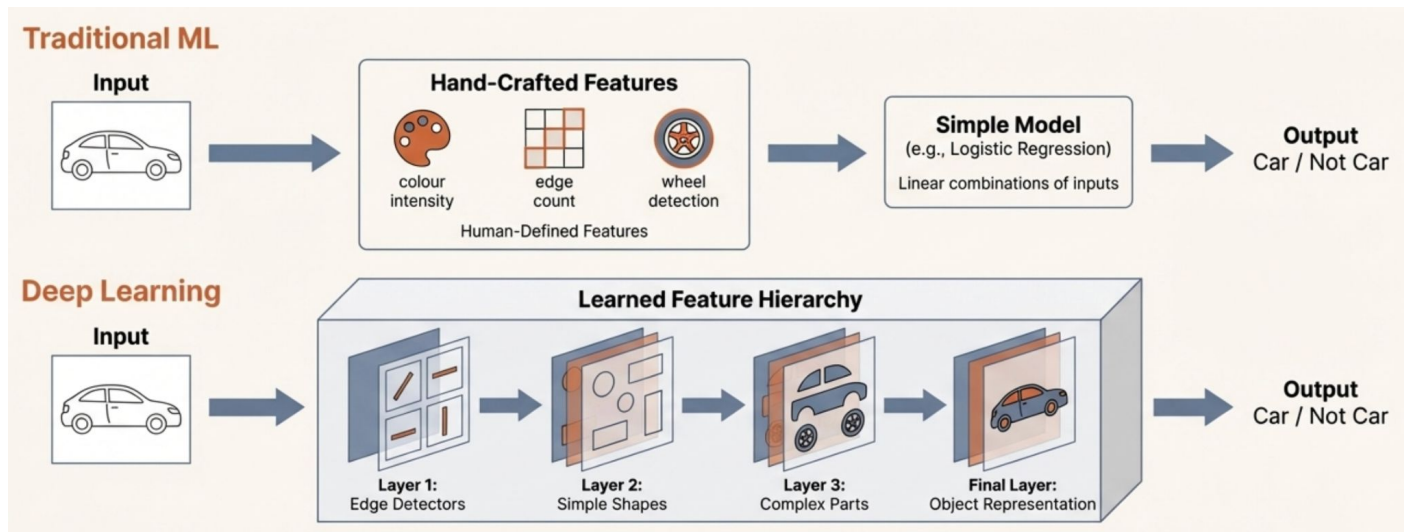
The Shift: Now, we use that foundation to build models that can generate, predict, and perceive.

The Connection: The mathematics you mastered—Bayes' Theorem, distributions, calculus—is the engine under the hood of everything we will build.

Where are we in the Map? (The sem's AI track)

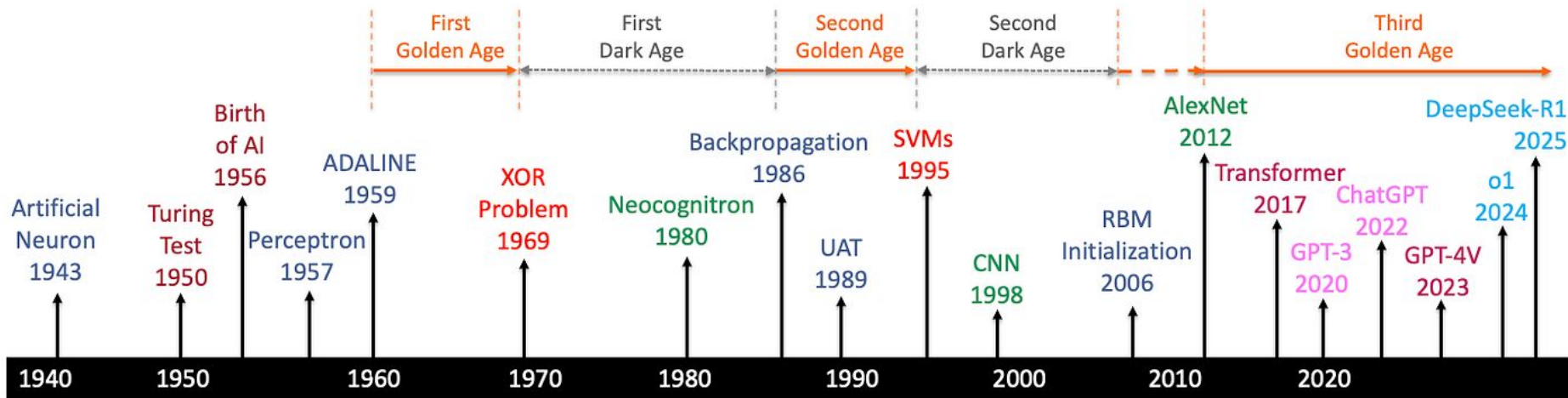


What is Deep Learning? (Representation Learning)

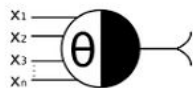


- **The Core Distinction:** In your Intro ML course, you often curated features (e.g., "color intensity", "roundness"). You told the machine what to look for.
- **Representation Learning:** In DL, we don't design features; we design architectures that learn Features.
- **Hierarchy:** The model automatically discovers that edges form textures, textures form shapes, and shapes form objects.
- **Key Insight:** We are replacing human heuristics with mathematical optimization.

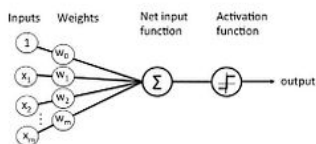
A Brief History of AI with Deep Learning



McCulloch-Pitts



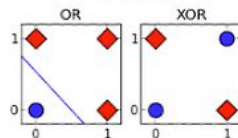
Rosenblatt



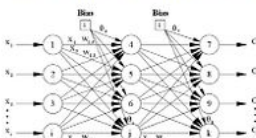
Widrow-Hoff



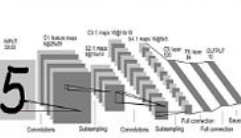
Minsky-Papert



Rumelhart, Hinton et al.



LeCun



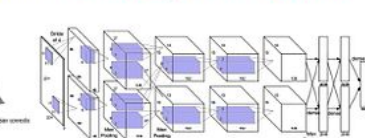
Hinton-Ruslan



Krizhevsky et al.



Vaswani



A (Sad) Story...



The Unsung Architect

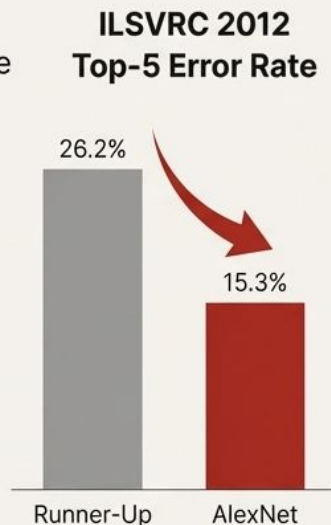
A Story of Innovation, Credit, and Anonymity at the Dawn of Deep Learning

An Excursion in the Deep Learning Course | Newton School of Technology

2012: The Year AI Had Its 'Big Bang'

The ImageNet Large Scale Visual Recognition Challenge (ILSVRC) global benchmark for computer vision.

- Before 2012: Progress was steady but incremental.
- In 2012: A new model, "AlexNet," shattered all records.



This wasn't just a win; it was a paradigm shift.

“It was a ‘sending a man to the moon’ moment for AI. Nothing was the same after.”

The Celebrated Pioneers

The AlexNet breakthrough is rightly credited to a team from the University of Toronto. Two of its authors have become titans of the industry, shaping the public conversation and future of AI.

- **Geoffrey Hinton:** The “Godfather of AI.” Continued his influential research, becoming a leading public voice on AI progress and safety.
- **Ilya Sutskever:** Co-founder and former Chief Scientist of **OpenAI**, the driving force behind models like GPT-4.



But the Paper is Titled 'AlexNet'...

Who is Alex?

Alex Krizhevsky
University of Toronto
kriz@cs.utoronto.ca

Ilya Sutskever
University of Toronto
ilyasu@cs.utoronto.ca

Geoffrey E. Hinton
University of Toronto
hinton@cs.utoronto.ca

Alexnet Paper

ImageNet Classification with Deep Convolutional Neural Networks



Abstract

We trained a large, deep convolutional neural network to classify the 1.2 million high-resolution images in the ImageNet LSVRC-2010 contest into the 1000 different classes. On the test data, we achieved top-1 and top-5 error rates of 37.5% and 17.0% which is considerably better than the previous state-of-the-art. The neural network, which has 60 million parameters and 650,000 neurons, consists of five convolutional layers, some of which are followed by max-pooling layers, and three fully-connected layers with a final 1000-way softmax. To make training faster, we used non-saturating neurons and a very efficient GPU implementation of the convolution operation. To reduce overfitting in the fully-connected layers we employed a recently-developed regularization method called “dropout” that proved to be very effective. We also entered a variant of this model in the ILSVRC-2012 competition and achieved a winning top-5 test error rate of 15.3%, compared to 26.2% achieved by the second-best entry.

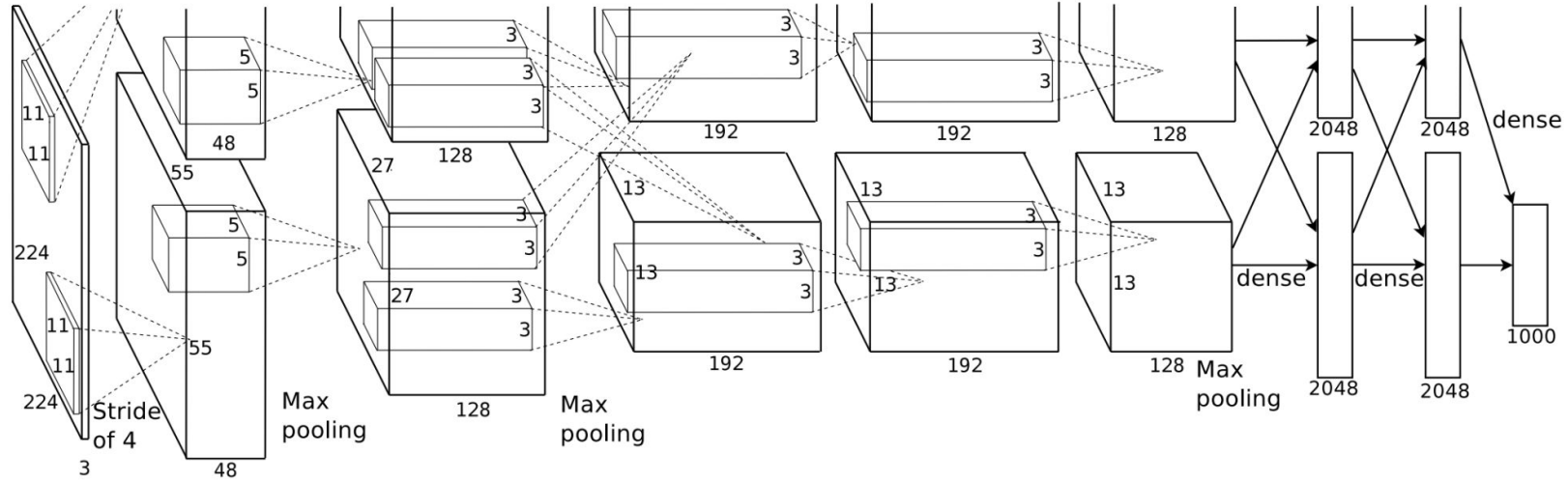


Figure 2: An illustration of the architecture of our CNN, explicitly showing the delineation of responsibilities between the two GPUs. One GPU runs the layer-parts at the top of the figure while the other runs the layer-parts at the bottom. The GPUs communicate only at certain layers. The network's input is 150,528-dimensional, and the number of neurons in the network's remaining layers is given by 253,440–186,624–64,896–64,896–43,264–4096–4096–1000.

Alex Krizhevsky: The First Author

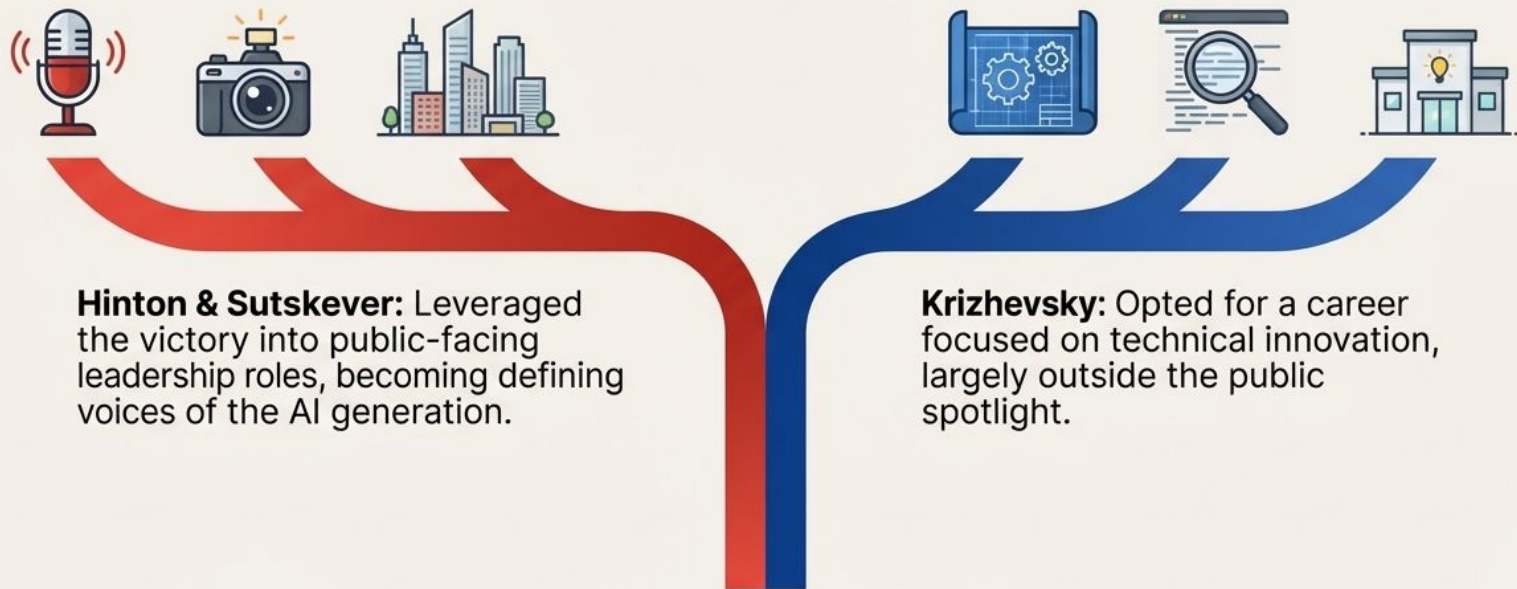


Alex Krizhevsky was the Ph.D. student who was the primary architect and implementer of the model that now bears his name.

- He was the first author on the groundbreaking paper, “ImageNet Classification with Deep Convolutional Neural Networks.”
- He developed the highly efficient GPU implementation that was critical for training such a large network, a massive engineering feat at the time.
- The architecture itself, which won the competition, is named AlexNet in his honor.

A Shared Victory, A Divergent Future

Following the **ImageNet** victory, the startup founded by the three researchers, **DNN Research Inc.**, was acquired by **Google**. This was the moment their paths began to diverge dramatically.



DNN Research Inc. acquired by **Google**

The Divergence: Public Persona vs. Quiet Innovation

The Public Persona

Ilya Sutskever

Co-founded OpenAI in 2015, becoming a central figure in the race for AGI. Later co-founded Safe Superintelligence Inc. in 2024.



Geoffrey Hinton

Continued influential research and became a **media staple** for his views on AI risk, earning the title "Godfather of AI".

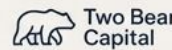
The Quiet Innovator

Alex Krizhevsky

Remained at Google until 2017.



Joined smaller, less-public firms like Dessa and later Two Bear Capital.



Focused primarily on technical innovation rather than media engagement or building a public brand.

The Anonymity of the Architect



Alex Krizhevsky

Key Question: Why does the person who engineered the revolution and gave the model his name remain largely unknown to the public?

- **Choice:** A deliberate preference for deep technical work over the demands of management and public life.
- **Personality:** Not all brilliant engineers are driven to be public figures or seek the spotlight.
- **The Nature of Credit:** The media and the market often simplify complex collaborations into a narrative of a few "Great Men," leaving the hands-on builders in the shadows.

All

Shorts

Videos

Unwatched

Watched

Recently uploaded

Live

Subscriptions

Under 4 min

4 - 20 min

Over 20 min

Filters

⋮



Ilya Sutskever, U of T honorary degree recipient, June 6, 2025

158K views • 6 months ago



Meet #UofT honorary degree recipient Ilya Sutskever, recognized for his excellence in the academy and outstanding service to the ...

CC

Ilya Sutskever's U of T honorary degree ceremony features a captivating speech. The AI visionary reflects on... **Summary** ▼

Geoffrey Hinton in conversation with Fei-Fei Li — Responsible AI development

209K views • 1 year ago



Recorded October 4, 2023. U of T Professor Emeritus Geoffrey Hinton is joined by Stanford University Professor Fei-Fei Li for a ...

CC

A conversation between two AI experts explores the ethical implications of AI development and its potential to... **Summary** ▼

Ilya Sutskever Calls for Superalignment Before Data Centers Evolve Into Artificial Life

79K views • 7 months ago



Ilyasutskever #ai #technology About Ilya Sutskever: Ilya Sutskever FRS is a Canadian-Israeli-Russian computer scientist ...

**3 chapters** Superalignment | Prosocial AI | Quantum Physics ▼

Meet Geoffrey Hinton, U of T's Godfather of Deep Learning

23K views • 8 years ago



What is Success?

Alex Krizhevsky continues to work on challenging technical problems, contributing to the field he helped create. His story forces us to ask fundamental questions about our own careers:

- Is success measured by **public acclaim**, or by the **private impact** of your work?
- Is it better to be the **face of a revolution**, or the one **building its engine**?

There is no right answer.

The (Bitter) Lesson...

A Lesson Written in 70 Years of AI Research.

Understanding the one pattern that has
consistently defined breakthrough progress.

AI research has historically faced a fundamental choice between two paths.



The Path of Human Knowledge

Leverage our own understanding of a domain.

Build in knowledge about the special structure of a problem.

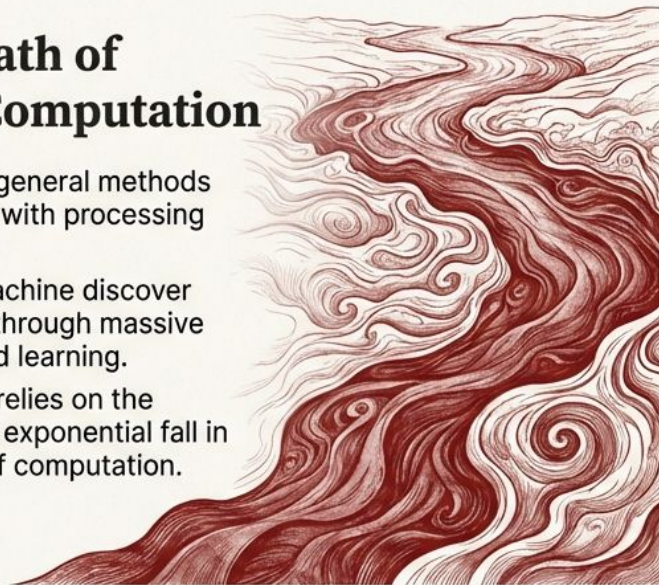
This approach is intuitive and personally satisfying for the researcher.

The Path of Raw Computation

Leverage general methods that scale with processing power.

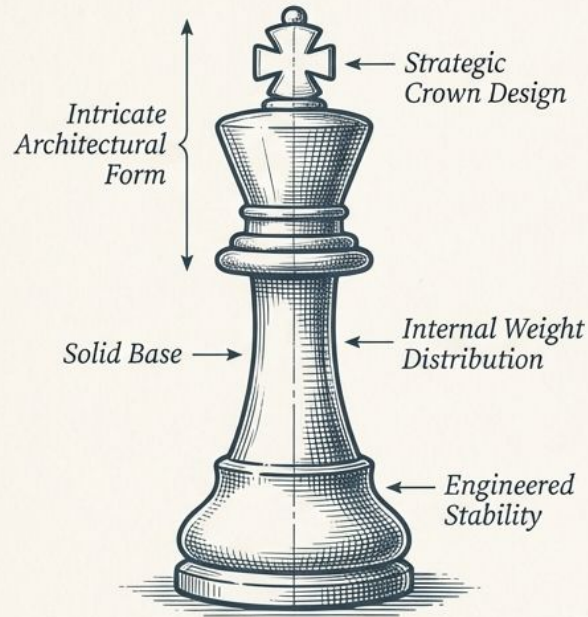
Let the machine discover solutions through massive search and learning.

This path relies on the continued exponential fall in the cost of computation.



"Most AI research has been conducted as if the computation available to the agent were constant... but, over a slightly longer time than a typical research project, massively more computation inevitably becomes available."

Case Study 1: In Computer Chess, the field first backed human-centric methods.



The majority of computer-chess researchers pursued methods that encoded human understanding of the game. They focused on building in knowledge of openings, endgames, and positional strategy. The goal was to create an elegant system that “thought” like a grandmaster, requiring less brute-force calculation.



But in 1997, massive, deep search defeated the world champion.



The methods that defeated Garry Kasparov were based on “massive, deep search.” This “brute force” approach, powered by special hardware, proved vastly more effective than the more elegant, human-knowledge-based systems.

Key Insight: This victory was met with dismay. Researchers who had invested in human-centric approaches “were not good losers.” They argued it wasn’t a general strategy and wasn’t “how people played chess.”

1970

1980

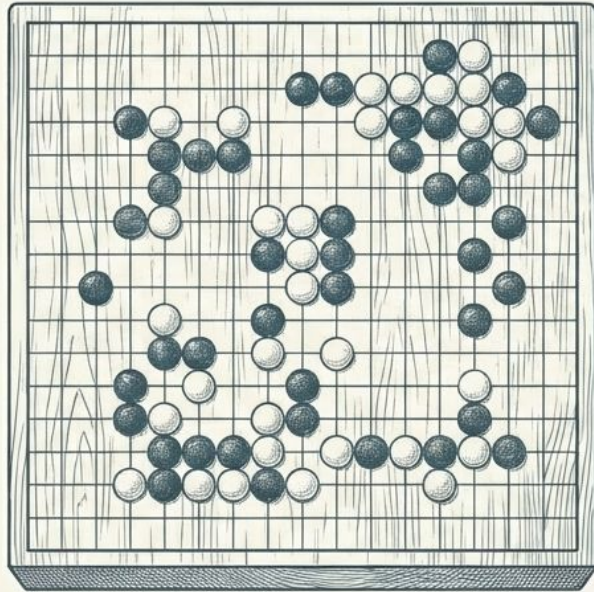
1990

2000

2010

2020

Case Study 2: Computer Go saw an identical pattern, delayed by 20 years.



For decades, the immense complexity of Go led to a consensus that pure search was infeasible. Initial efforts focused heavily on leveraging human knowledge of the game's special features to avoid massive search.



Key Insight:

"All those efforts proved irrelevant, or worse, once search was applied effectively at scale."

1970

1980

1990

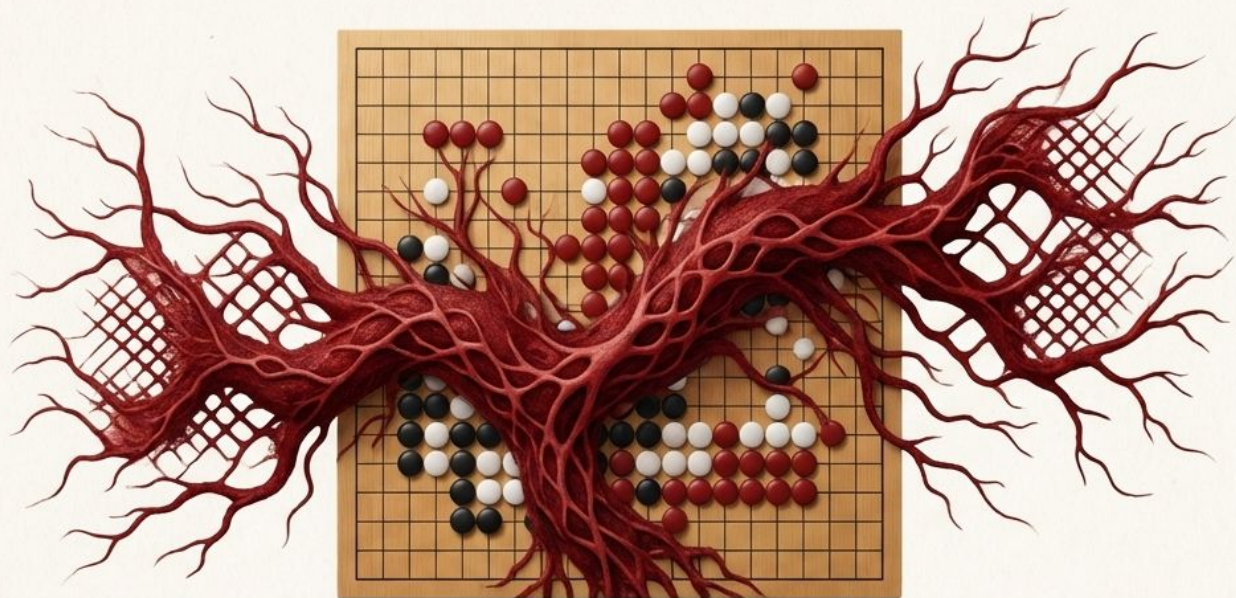
2000

2010

2016

2020

The breakthrough in Go combined massive search with learning from self-play.

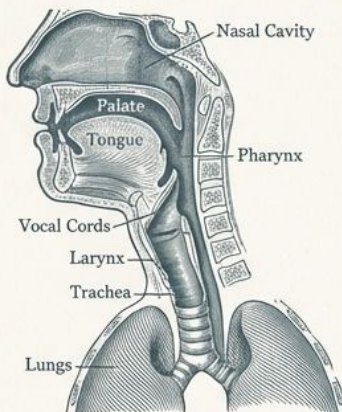


The ultimate success came from embracing search and, critically, learning. Learning by self-play allowed the system to generate its own knowledge, creating a powerful value function without human input.

"Search and learning are the two most important classes of techniques for utilizing massive amounts of computation in AI research."

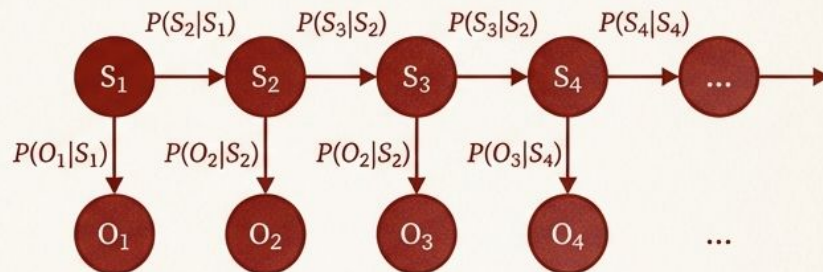
Case Study 3: Speech recognition followed the same arc from knowledge to statistics.

Human Knowledge



Early methods were based on explicit knowledge of words, phonemes, and the physical structure of the human vocal tract.

Computation & Statistics



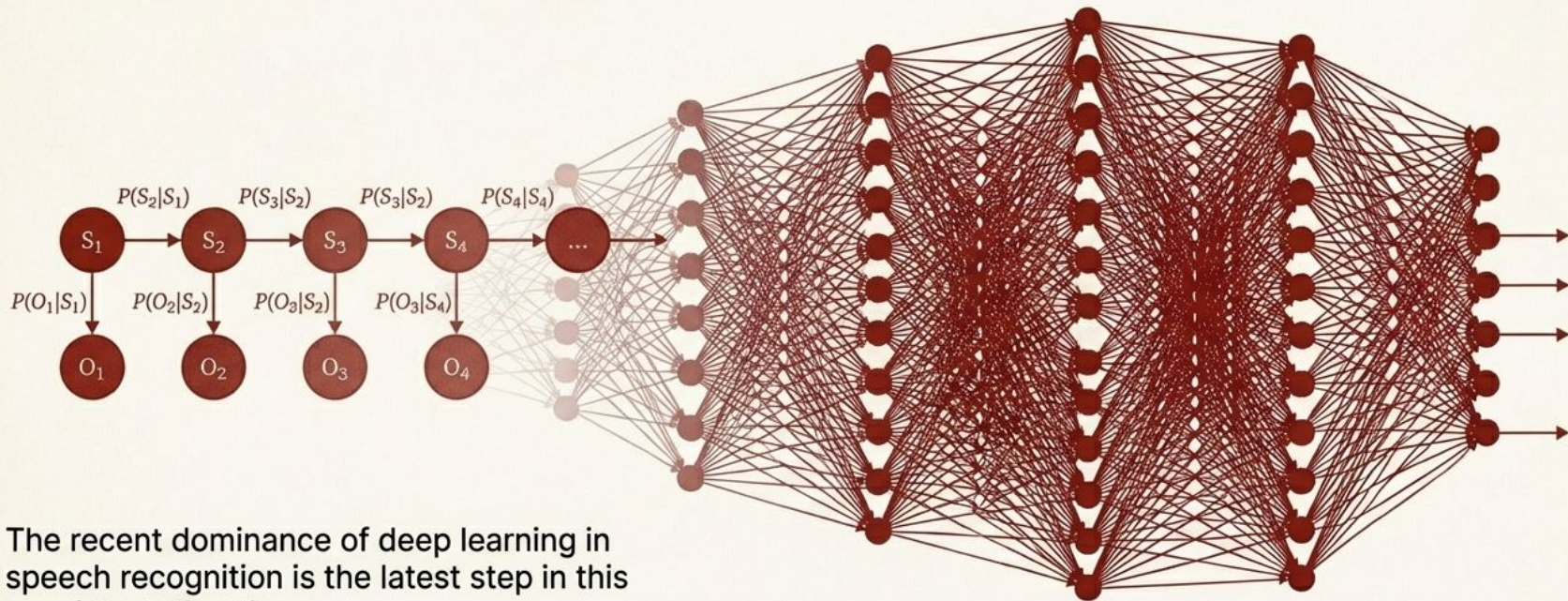
Newer methods based on HMMs were more statistical, leveraged more computation, and quickly proved superior.

💡 Key Insight:

"Again, the statistical methods won out over the human-knowledge-based methods." This victory triggered a gradual but major shift across all of natural language processing.



Today, deep learning has pushed this trend to its logical conclusion.

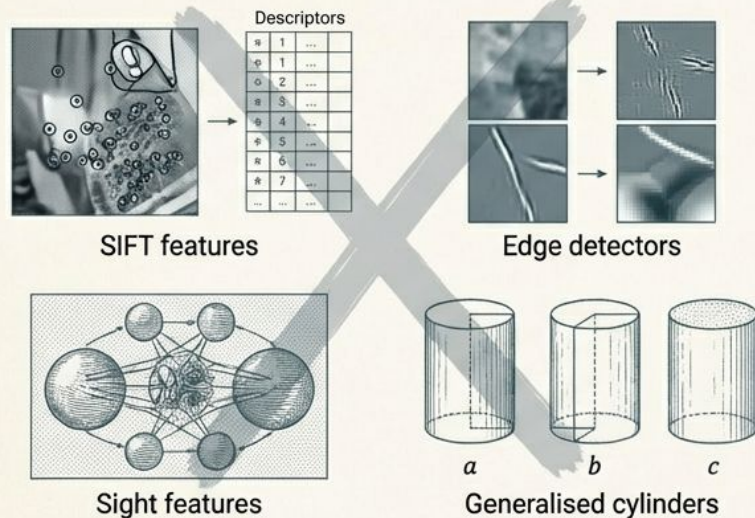


The recent dominance of deep learning in speech recognition is the latest step in this consistent direction.

“Modern deep learning methods “rely even less on human knowledge, and use even more computation, together with learning on huge training sets, to produce dramatically better speech recognition systems.”

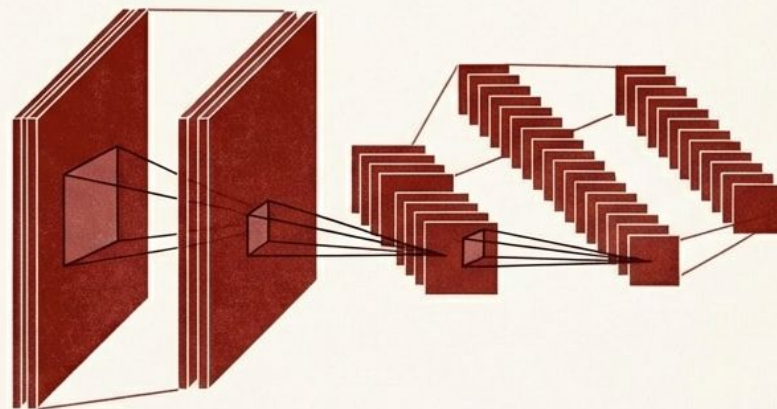
Case Study 4: Computer vision shows the same shift away from engineered features.

Discarded Human-Engineered Features



Early methods were built on concepts researchers thought were fundamental to sight, such as edges or specific shapes.

Modern Learned Representations



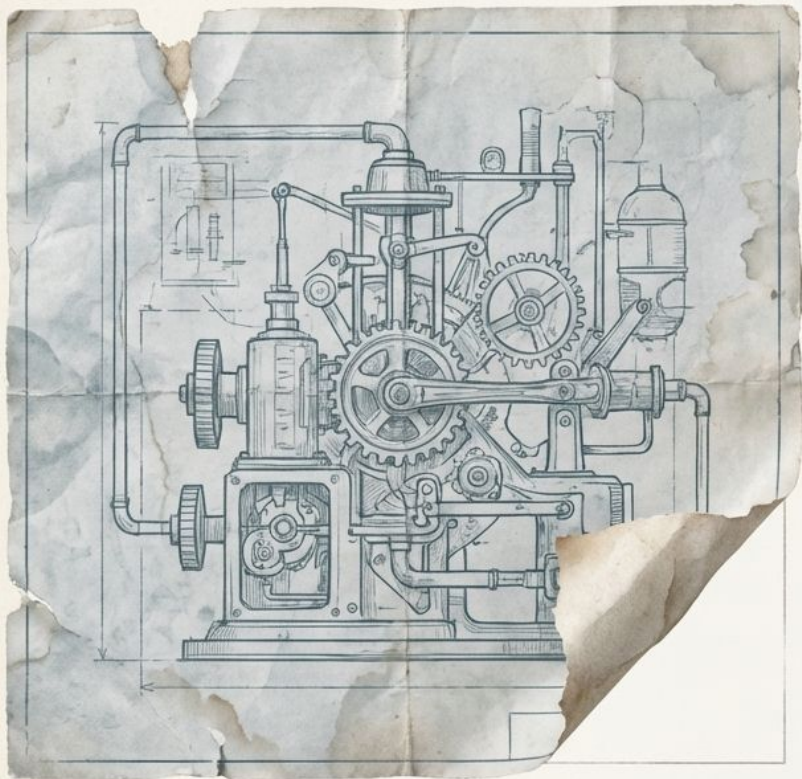
Modern deep-learning networks use only the notions of convolution and certain kinds of invariances, and perform much better.

The Bitter Lesson

The biggest lesson that can be read from 70 years of AI research is that general methods that leverage computation are ultimately the most effective, and by a large margin.

- Researchers often try to build their own knowledge into agents.
- This always helps in the short term and is personally satisfying.
- In the long run, this approach plateaus and even inhibits further progress.
 - Breakthrough progress eventually arrives from an opposing approach: scaling computation with search and learning.

Why is the lesson ‘bitter’?



The success of computation-scaled methods is often “tinged with bitterness, and often incompletely digested.”

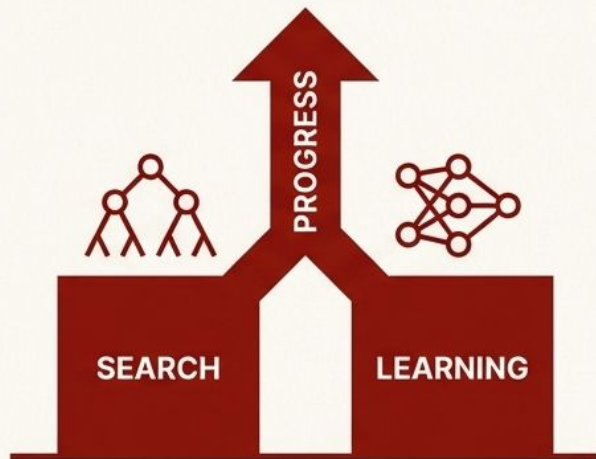
It is success over a favored, human-centric approach. Researchers wanted methods based on their own clever insights to win.

There is a psychological commitment to the idea that building in how we to *think* we think should be the superior path.

The lesson is a humbling one.

Implication 1: The future belongs to general-purpose methods that scale.

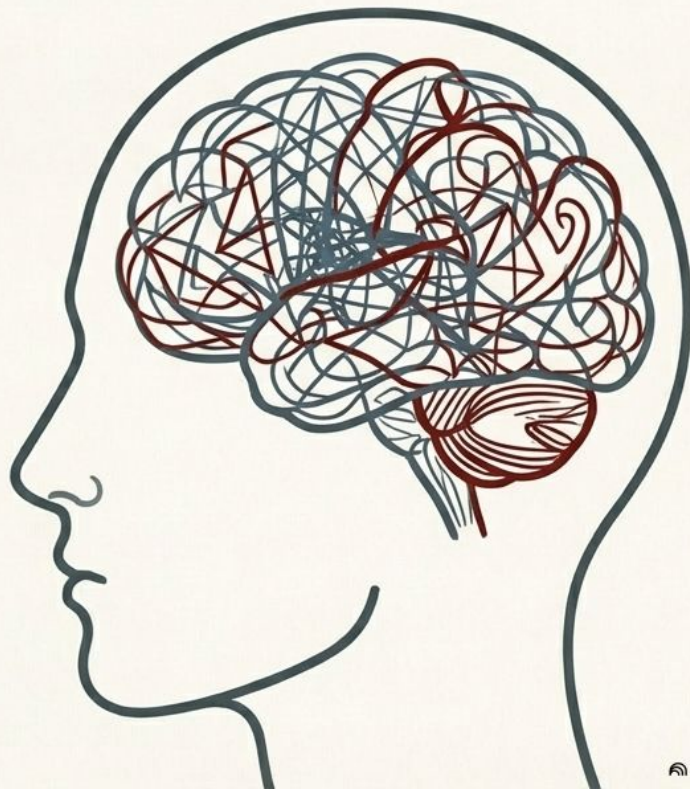
The core takeaway is to recognise and embrace the “great power of general purpose methods.” We must prioritise methods that continue to improve as available computation becomes very great.



Implication 2: We must stop trying to build in the contents of minds.

“The actual contents of minds are tremendously, irredeemably complex.” We should stop trying to find simple ways to represent complex concepts like space, objects, or symmetries and build them into our systems.

“Their complexity is endless.” Building in our own simplified models of the world only makes it harder for the agent to learn the true, arbitrary complexity of reality.



The goal is not to build agents that contain our knowledge, but agents that can discover for themselves.

Instead of hard-coding our discoveries, we should build in the **meta-methods** that enable discovery itself. The search for good approximations should be conducted by our methods, not by us.



*"We want AI agents that can discover like we can, not **which contain what we have discovered**. Building in our discoveries only makes it harder to see how the discovering process can be done."*

This course...

The Two Pillars: Math & Code

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial a} \frac{\partial a}{\partial z} \frac{\partial z}{\partial w}$$

The Math (The Physics): Deep Learning is not magic; it is Calculus and Linear Algebra acting on Data.

- We will open the “Black Box”. You will derive Backpropagation by hand.

- Why? Because when the model fails (and it will), the math tells you how to fix it.

```
loss.backward()
```

The Code (The Engineering): We build what we derive.

- Proficiency in PyTorch/TensorFlow is now a standard literacy.

- We don’t just call libraries; we implement the computational graph to verify our mathematical intuition.

Why take this course seriously? (The 3 Rationales)



The Foundation



The Utility



The Fascination

Rationale 1: The Degree (The Foundation)

- It is a B.Tech requirement, but more importantly:
- AI is no longer a “niche” elective. It is the new “Calculus”—a fundamental literacy required for modern engineering.
- You cannot claim to be a Computer Science graduate in 202X without understanding how a Neural Network updates its weights.



Rationale 2: The Job (The Utility)

Market Reality:

"Intro to ML" is no longer a differentiator on a resume. Everyone has it.

The Deep Learning Edge:

Industry demands specific skills—PyTorch/TensorFlow proficiency, ability to read research papers, experience with large models (LLMs/Vision), deployment and monitoring capabilities.

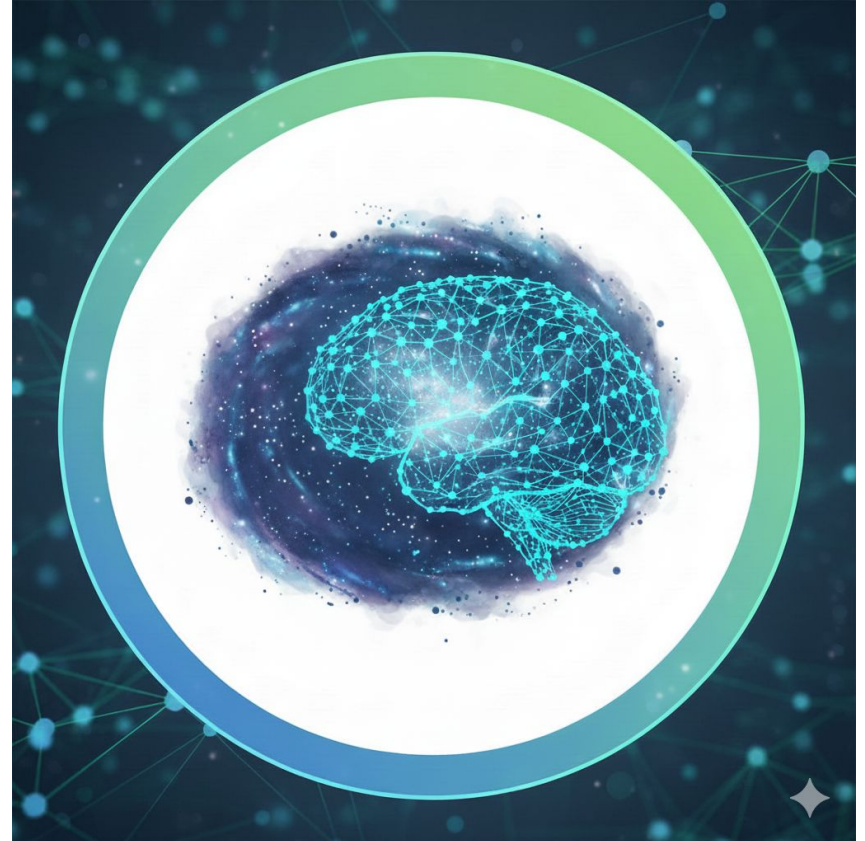
Roles:

ML Engineer, Research Scientist, Data Engineer (specialized)



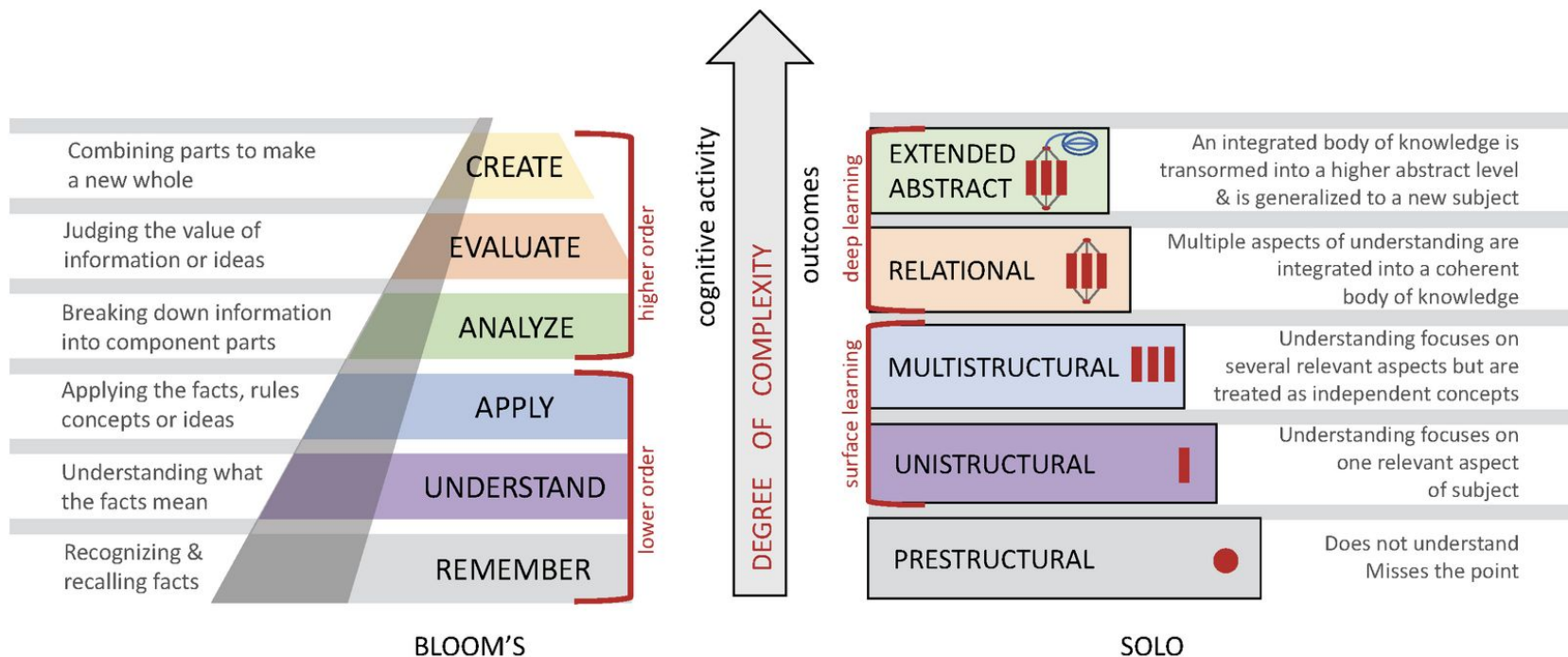
Rationale 3: The Fascination (The Spark)

- This is the most exciting field in human history right now.
- We are not just coding loops; we are **approximating intuition**.
- The **"Magic"**: We will learn how mathematical matrices can learn to "see" a cat, "write" poetry, or "fold" proteins.
- If you aren't curious about how ChatGPT works and much more, you are in the wrong room.



How will we learn?

BLOOM's & SOLO TAXONOMY HIERARCHIES

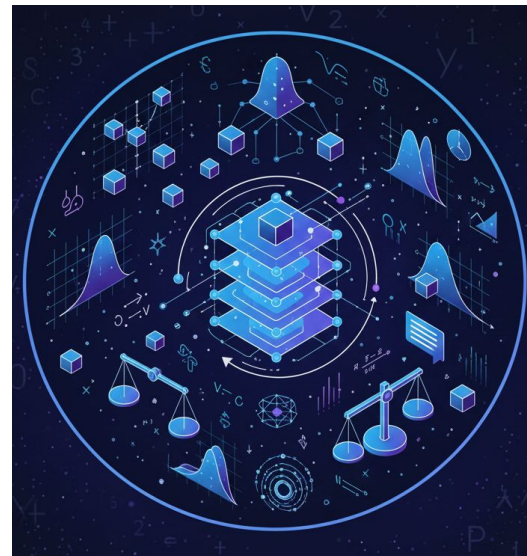


Course Architecture (The Syllabus)

Module 1 - Core Probability Theory



Linear Algebra – Foundations and Operations; Structure, Decomposition, and Applications



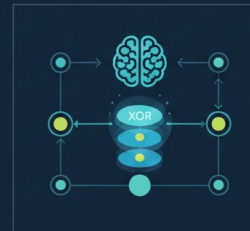
Probability Foundations for DL; Distributions, Bayes' Rule & Information Theory

Course Architecture (The Syllabus)

Module 2 - Feed Forward Networks

Deep Feedforward Networks Intro & XOR
Example; Hidden Units & Network
Architecture

Backpropagation & Training Mechanics;
Regularization for Deep Learning



DEEP FEEDFORWARD
NETWORKS



HIDDEN UNITS &
ARCHITECTURE



BACKPROPAGATION
& TRAINING



REGULARIZATION



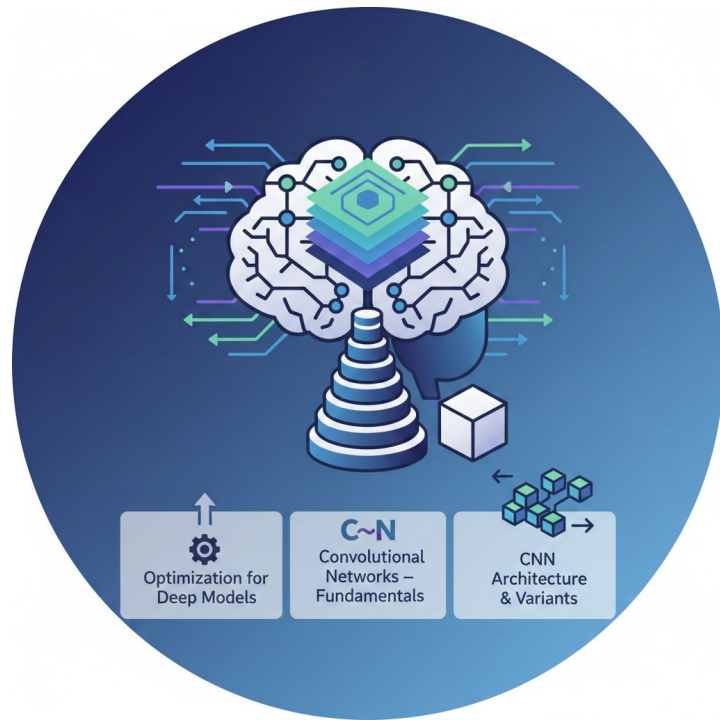
Course Architecture (The Syllabus)

Module 3 - Optimisation and CNNs

Optimization for Deep Models

Convolutional Networks – Fundamentals
CNN Architecture & Variants

Extensions & Neuroscience Inspiration



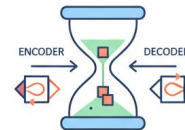
Course Architecture (The Syllabus)

Module 4 - Sequence Modeling

Sequence Modeling & Recurrent Networks;
Large-Scale DL & Computer Vision Application



Speech & NLP Applications; Autoencoders –
Foundations



Course Architecture (The Syllabus)

Module 5 - Autoencoders and Transformers

Autoencoders – Advanced Variants
& Applications; Transformers – The
Need for Attention

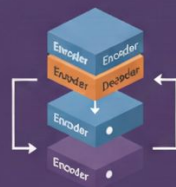
Transformer Architecture in Depth;
Large Language Models &
Applications



Autoencoders



Transformers - The
Need for Attention



Transformer
Architecture in Depth



Large Language Models
& Applications

The Shared Project (Joint Ops)



Evaluation Components

Function	Weightage
Attendance/Class Participation	5%
In Class/Post Class Assignments	10%
Mini-projects	25%
Mid Semester	20%
End Semester	40 %

Resources

[Deep Learning Resources](#)

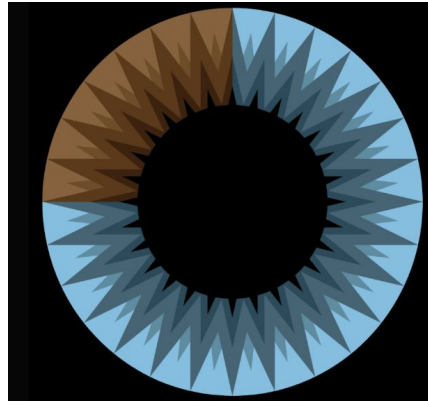
Reference Books

- Deep Learning By **Ian Goodfellow**, Yoshua Bengio, and Aaron Courville
- Pattern Recognition and Machine Learning By **Christopher M. Bishop**

Other References



Khan Academy



3Blue1Brown



deeplearning.ai

Final Thoughts

- This course is mathematically rigorous.
- It moves fast.
- By the end, you will understand the “ghost in the machine.”